

Multiple Choice (10 marks)

1. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
 - (a) Planck's constant
 - (b) Boltzmann's constant
 - (c) The Rydberg constant
 - (d) The speed of light
2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
3. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
4. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
5. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
 - (a) 0.50 c
 - (b) 0.60 c
 - (c) 0.70 c
 - (d) 0.80 c

True / False (10 marks)

Answer True or False

6. True or False: The emission spectrum of the doubly ionized lithium atom Li^{++} has wavelengths decreased by a factor of 9 compared to hydrogen.
7. True or False: The energy needed to ionize helium by removing one electron is 39.5 eV.
8. True or False: A meter stick moving at $0.8c$ takes 4.2 ns to pass the observer in their reference frame.
9. True or False: The sign of the charge carriers in a doped semiconductor can be deduced from measurements of thermal conductivity.
10. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen is 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the work done by an ideal monatomic gas during a quasi-static expansion to twice its volume in isothermal and adiabatic processes. Analyze the relationship between these two values.
12. Analyze a one-dimensional collision where a particle of mass $2m$ collides with a stationary particle of mass m , and they stick together. Discuss the fraction of initial kinetic energy lost in the process.

End of Paper